Chapter 5

Determinants

Homework List

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Sec 5.1

Properties of the determinant.

What is a determinant?

Definition:

- **The determinant of a square matrix is a single number.**
  - **If a matrix has a determinant of 0, it is singular; the inverse does not exist.**
  - **If the determinant is NOT zero, the matrix is nonsingular.**
- **Just as the matrix inverse is only defined for square matrices, the determinant is only defined for square matrices.**

The 2-by-2 case.

- **The determinant of a square matrix is a number (product of the diagonal numbers in the row echelon form) that can be used to determine whether or not the matrix has an inverse.**

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} a & b \\ 0 & d - \dfrac{bc}{a} \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & \dfrac{ad - bc}{a} \end{bmatrix}, a \neq 0$$

- **Inverse exists if and only if:**

$$ad - bc \neq 0$$

$$\det(A) = |A| = (ad - bc)$$

The 3-by-3 case.

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}-\dfrac{a_{12}a_{21}}{a_{11}} & a_{23}-\dfrac{a_{13}a_{21}}{a_{11}} \\ 0 & a_{32}-\dfrac{a_{12}a_{31}}{a_{11}} & a_{33}-\dfrac{a_{13}a_{31}}{a_{11}} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \dfrac{a_{22}a_{11}-a_{12}a_{21}}{a_{11}} & \dfrac{a_{23}a_{11}-a_{13}a_{21}}{a_{11}} \\ 0 & \dfrac{a_{32}a_{11}-a_{12}a_{31}}{a_{11}} & \dfrac{a_{33}a_{11}-a_{13}a_{31}}{a_{11}} \end{bmatrix}, a_{11} \neq 0$$

$$\rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \dfrac{a_{22}a_{11}-a_{12}a_{21}}{a_{11}} & \dfrac{a_{23}a_{11}-a_{13}a_{21}}{a_{11}} \\ 0 & 0 & \dfrac{a_{33}a_{11}-a_{13}a_{31}}{a_{11}} - \left(\dfrac{a_{23}a_{11}-a_{13}a_{21}}{a_{11}}\right)\left(\dfrac{a_{32}a_{11}-a_{12}a_{31}}{a_{22}a_{11}-a_{12}a_{21}}\right) \end{bmatrix}, a_{22}a_{11}-a_{12}a_{21} \neq 0$$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \dfrac{a_{22}a_{11}-a_{12}a_{21}}{a_{11}} & a_{23} \\ 0 & 0 & \dfrac{(a_{33}a_{11}-a_{13}a_{31})(a_{22}a_{11}-a_{12}a_{21}) - (a_{23}a_{11}-a_{13}a_{21})(a_{32}a_{11}-a_{12}a_{31})}{a_{11}(a_{22}a_{11}-a_{12}a_{21})} \end{bmatrix}$$

The 3-by-3 case.

$$\frac{(a_{33}a_{11}-a_{13}a_{31})(a_{22}a_{11}-a_{12}a_{21})-(a_{23}a_{11}-a_{13}a_{21})(a_{32}a_{11}-a_{12}a_{31})}{a_{11}(a_{22}a_{11}-a_{12}a_{21})}$$

$$\downarrow$$

$$\frac{a_{33}a_{11}a_{22}a_{11}-a_{33}a_{11}a_{12}a_{21}-a_{13}a_{31}a_{22}a_{11}+a_{13}a_{31}a_{12}a_{21}-a_{23}a_{11}a_{32}a_{11}+a_{23}a_{11}a_{12}a_{31}+a_{13}a_{21}a_{32}a_{11}-a_{13}a_{21}a_{12}a_{31}}{a_{11}(a_{22}a_{11}-a_{12}a_{21})}$$

$$\downarrow$$

$$\frac{a_{33}a_{11}a_{22}a_{11}-a_{33}a_{11}a_{12}a_{21}-a_{13}a_{31}a_{22}a_{11}-a_{23}a_{11}a_{32}a_{11}+a_{23}a_{11}a_{12}a_{31}+a_{13}a_{21}a_{32}a_{11}}{a_{11}(a_{22}a_{11}-a_{12}a_{21})}$$

$$\downarrow$$

$$\frac{a_{33}a_{11}a_{22}-a_{33}a_{12}a_{21}-a_{13}a_{31}a_{22}-a_{23}a_{32}a_{11}+a_{23}a_{12}a_{31}+a_{13}a_{21}a_{32}}{(a_{22}a_{11}-a_{12}a_{21})}$$

- **Inverse exists if and only if:**

$$a_{33}a_{11}a_{22}-a_{33}a_{12}a_{21}-a_{13}a_{31}a_{22}-a_{23}a_{32}a_{11}+a_{23}a_{12}a_{31}+a_{13}a_{21}a_{32}\neq 0$$

$$\det(A)=|A|=a_{11}a_{22}a_{33}-a_{11}a_{23}a_{32}+a_{12}a_{23}a_{31}-a_{12}a_{21}a_{33}+a_{13}a_{21}a_{32}-a_{13}a_{22}a_{31}$$

Determinant of the triangular matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & & a_{3n} \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & a_{nn} \end{bmatrix}$$

- **Inverse exists if and only if all diagonal entries are non-zero.**

$$a_{11}, a_{22}, a_{33}, \cdots, a_{nn} \neq 0$$

$$\det(A) = |A| = a_{11}a_{22}a_{33}\cdots a_{nn} = \sum_{i=1}^{n} a_{ii}$$

Properties of the determinant.

- **The determinant of the identity matrix is one:**

$$\det(I) = |I| = 1$$

- **The determinant of any singular matrix is zero:**

$$\det(A) = |A| = 0 \Leftrightarrow \nexists A^{-1}$$

Singular matrix examples:

- **Any matrix with identical rows.**
- **Any matrix with identical columns.**
- **Any matrix with a zero row or a zero column.**

Properties of the determinant

- **Exchanging rows changes the sign of the determinant:**

$$A \xrightarrow{R_i \leftrightarrow R_j} B \Rightarrow |B| = -|A|$$

- **Scaling a row scales the determinant by the same amount:**

$$A \xrightarrow{kR_i \to R_i} B \Rightarrow |B| = k|A|$$

- **Adding a scalar multiple of one row to another does not change the determinant :**

$$A \xrightarrow{R_i + kR_j \to R_i} B \Rightarrow |B| = |A|$$

Examples:

Check: $\begin{vmatrix} c & d \\ a & b \end{vmatrix} = -\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ (both sides equal $bc - ad$).

multiply row 1 by any number t
det is multiplied by t

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

add row 1 of A to row 1 of A':
then determinants add

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}.$$

ℓ times row 1
from row 2

$$\begin{vmatrix} a & b \\ c-\ell a & d-\ell b \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

Examples

Row of zeros $\begin{vmatrix} 0 & 0 \\ c & d \end{vmatrix} = 0$ and $\begin{vmatrix} a & b \\ 0 & 0 \end{vmatrix} = 0.$

Triangular $\begin{vmatrix} a & b \\ 0 & d \end{vmatrix} = ad$ and also $\begin{vmatrix} a & 0 \\ c & d \end{vmatrix} = ad.$

Properties of the determinant:

- **The determinant of a product of matrices is the product of the determinants:**

$$C = AB \Rightarrow |C| = |A||B|$$

- **The determinant of the inverse is the inverse of the determinant:**

$$\left|A^{-1}\right| = (|A|)^{-1} = \frac{1}{|A|}$$

- **A matrix has the same determinant as its transpose:**

$$|A| = \left|A^T\right|$$

9 ***The determinant of AB is* det A *times* det B:** $|AB| = |A|\,|B|$.

Product rule

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} p & q \\ r & s \end{vmatrix} = \begin{vmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{vmatrix}.$$

This product rule is the most intricate so far. Even the 2 by 2 case needs some algebra:

$$|A|\,|B| = (ad - bc)(ps - qr) = (ap + br)(cq + ds) - (aq + bs)(cp + dr) = |AB|.$$

For the n by n case, here is a snappy proof that $|AB| = |A|\,|B|$. When $|B|$ is not zero, consider the ratio $\boldsymbol{D(A) = |AB|/|B|}$. *Check that this ratio* $D(A)$ *has properties* 1,2,3. Then $D(A)$ has to be the determinant and we have $|AB|/|B| = |A|$. Good.

Note: If B is singular, then |B| =0. From previous sessions, we know that AB is singular if A or B is singular.

Determinants and elimination.

- **Consider Gaussian elimination as represented by the LU decomposition:**

$$A \xrightarrow{P} A' \xrightarrow{L^{-1}} U, \qquad PA = LU$$

$$|P||A| = |L||U| \rightarrow |A| = \frac{|L||U|}{|P|}$$

- P **is an identity matrix with rearranged rows, and** L **is a lower triangular matrix with ones on the diagonal:**

$$|P| = \pm 1, |L| = 1 \ \rightarrow |A| = \pm |U|$$

Determinants and elimination.

The determinant of any matrix can be found using elimination:

- **Perform forward elimination, without row scaling, to an echelon form:**

$$A \rightarrow U$$

- **The product of diagonal entries in echelon form gives determinant of the original matrix to within a sign ($\pm$ product of pivots):**

$$|A| = \pm|U|$$

- **Count the row exchanges: if there were an even number, leave the sign unchanged, if there were an odd number change the sign.**

Determinants and elimination.

Example:

$$\begin{vmatrix} 1 & 2 & 1 & 2 \\ 2 & 2 & 2 & 1 \\ 1 & 0 & 1 & -2 \\ -1 & 2 & 2 & 4 \end{vmatrix} = ?$$

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 2 & 2 & 1 \\ 1 & 0 & 1 & -2 \\ -1 & 2 & 2 & 4 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -2 & 0 & -3 \\ 0 & -2 & 0 & -4 \\ 0 & 4 & 3 & 6 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -2 & 0 & -3 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 3 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -2 & 0 & -3 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = U$$

- **We used one row exchange (odd):**

$$|U| = (1)(-2)(3)(-1) = +6, \qquad |A| = -|U| = -6$$

Examples

- Properties of determinant.

Ex 1: Find the determinant

Find the determinant:

$$|L|, \qquad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 \\ 4 & -2 & 4 & 0 \\ 5 & -3 & 5 & -3 \end{bmatrix}$$

Find the determinant

$$L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 \\ 4 & -2 & 4 & 0 \\ 5 & -3 & 5 & -3 \end{bmatrix}$$

- **Lower triangular matrix, simply take the product of the diagonal elements:**

$$|L| = (2)(-1)(4)(-3) = 24$$

Ex 2: Find the determinant

Find the determinant:

$$|P|, \qquad P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Find the determinant

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Three (odd) row exchanges to get the identity matrix:**

$$|P| = -|I| = -1$$

Ex 3: Find the determinant

Find the determinant:

$$|A|, \qquad A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

Find the determinant

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & -1/2 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & -1/2 & 3 \\ 0 & 0 & 0 & 16 \end{bmatrix} = U$$

- **Elimination without row exchange:**

$$|A| = |U| = (1)(-2)\left(\frac{-1}{2}\right)(16) = 16$$

Ex 4: Find the determinant

Find the determinant:

$$|A|, \qquad A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 1 \\ 2 & 1 & 2 & 3 \end{bmatrix}$$

Find the determinant

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 1 \\ 2 & 1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1/2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1/2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = U$$

- **We used two row exchanges (even):**

$$|U| = (1)(2)\left(\frac{1}{2}\right)(1) = 1, \qquad |A| = |U| = 1$$

Sec 5.2

Permutations and Cofactors

Direct forms for the determinant.

- **The determinant of any matrix can be found using elimination.**
- **Are there more direct definitions?**
- **2x2:** $$|A| = (a_{11}a_{22} - a_{12}a_{21})$$
- **3x3:** $$\begin{aligned} |A| = \quad & +a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ & -a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \end{aligned}$$
- **Sum of terms, each a product of n elements. First indices follow 1, 2, … , n; second indices are "scrambled."**

Leibniz formula: Permutations

Formal definition:

$$\det(A) = |A| = \sum_{\sigma \in S_n} \left(\operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i\sigma(i)} \right)$$

- S_n: **The set of all possible permutations of the sequence 1, 2, … n.**
- σ: **One particular permutation (re-ordering) of the sequence 1, 2, …n.**
- $\sigma(i)$: **The ith element of a particular permutation.**
- $\operatorname{sgn}(\sigma)$: **The *parity* of the permutation.** $\operatorname{sgn}(\sigma) = (-1)^p$ **where** p **is number of pairwise swaps to required to convert the permutation into the original sequence.**

Elegant, but not practical, especially for "by-hand" calculations.

Example

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & & \\ & a_{22} & \\ & & a_{33} \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ & & a_{23} \\ a_{31} & & \end{vmatrix} + \begin{vmatrix} & & a_{13} \\ a_{21} & & \\ & a_{32} & \end{vmatrix}$$

Six terms

$$+ \begin{vmatrix} a_{11} & & \\ & & a_{23} \\ & a_{32} & \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ a_{21} & & \\ & & a_{33} \end{vmatrix} + \begin{vmatrix} & & a_{13} \\ & a_{22} & \\ a_{31} & & \end{vmatrix}.$$

There are $3! = 6$ ***ways to order the columns, so six determinants***. The six permutations of $(1, 2, 3)$ include the identity permutation $(1, 2, 3)$ from $P = I$.

$$\textbf{Column numbers} = (1,2,3), (2,3,1), (\mathbf{3,1,2}), (1,3,2), (2,1,3), (3,2,1). \qquad (6)$$

Example: Continued

$$\det A = a_{11}a_{22}a_{33}\begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix} + a_{12}a_{23}a_{31}\begin{vmatrix} & 1 & \\ & & 1 \\ 1 & & \end{vmatrix} + a_{13}a_{21}a_{32}\begin{vmatrix} & & 1 \\ 1 & & \\ & 1 & \end{vmatrix}$$
$$+ a_{11}a_{23}a_{32}\begin{vmatrix} 1 & & \\ & & 1 \\ & 1 & \end{vmatrix} + a_{12}a_{21}a_{33}\begin{vmatrix} & 1 & \\ 1 & & \\ & & 1 \end{vmatrix} + a_{13}a_{22}a_{31}\begin{vmatrix} & & 1 \\ & 1 & \\ 1 & & \end{vmatrix}. \quad (7)$$

The first three (even) permutations have $\det P = +1$, the last three (odd) permutations have $\det P = -1$. We have proved the 3 by 3 formula in a systematic way.

Facts

1 **2 by 2**: $ad - bc$ has $2!$ terms with $\pm$ signs. **n by n: det A adds $n!$ terms with $\pm$ signs.**

2 For $n = 3$, $\det A$ adds $3! = 6$ terms. Two terms are $+a_{12}a_{23}a_{31}$ and $-a_{13}a_{22}a_{31}$. **Rows 1, 2, 3 and columns 1, 2, 3 appear once in each term.**

3 That minus sign came because the column order $3, 2, 1$ needs one exchange to recover $1, 2, 3$.

Strang's Book: Big Formula

$$\det A = \text{sum over all } \mathbf{n}! \text{ column permutations } P = (\alpha, \beta, \ldots, \omega)$$

$$= \boxed{\sum (\det P) a_{1\alpha} a_{2\beta} \cdots a_{n\omega}} = \textbf{BIG FORMULA}. \tag{8}$$

Laplace expansion: Cofactors (Useful!)

Expansion on row i:

$$|A| = \sum_{j=1}^{n} a_{ij} C_{ij}\,, \qquad C_{ij} = (-1)^{i+j} |M_{ij}|$$

Expansion on column j:

$$|A| = \sum_{i=1}^{n} a_{ij} C_{ij}\,, \qquad C_{ij} = (-1)^{i+j} |M_{ij}|$$

- M_{ij} **is the matrix obtained when row i and column j are removed from** A**.**
- $|M_{ij}|$ **is called the ijth minor of** A**.**
- C_{ij} **is the called the ijth cofactor of** A**.**

Direct forms for the determinant.

Note: The sign || means the determinant, not the absolute value.

- **2x2:** $|A| = (a_{11}a_{22} - a_{12}a_{21})$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}|a_{22}| - a_{12}|a_{21}|$$

- **3x3:**

$$|A| = \begin{aligned} &+a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ &-a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \end{aligned}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- **A sum of n determinants of (n-1)-by-(n-1) matrices.**

Direct forms for the determinant.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \rightarrow \begin{vmatrix} a_{11} & & \\ & a_{22} & a_{23} \\ & a_{32} & a_{33} \end{vmatrix}, \begin{vmatrix} & a_{12} & \\ a_{21} & & a_{23} \\ a_{31} & & a_{33} \end{vmatrix}, \begin{vmatrix} & & a_{13} \\ a_{21} & a_{22} & \\ a_{31} & a_{32} & \end{vmatrix}$$

- **Each term includes one element of the first row, and the determinant of the matrix left after deleting the row and column of the coefficient.**

Recursive use of Laplace expansion

$$\begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 2 & 1 & 2 & 0 & 0 \\ 0 & 2 & 1 & 2 & 0 \\ 0 & 0 & 2 & 1 & 2 \\ 0 & 0 & 0 & 2 & 1 \end{vmatrix} = 1 \begin{vmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{vmatrix}$$

$$= 1\left(1\begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix} - 2\begin{vmatrix} 2 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix}\right) - 2\left(2\begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix}\right) = -3\begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix} - 2\begin{vmatrix} 2 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix}$$

$$= -3\left(1\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} - 2\begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix}\right) - 2\left(2\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}\right) = -7\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} + 6\begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix}$$

$$= -7(1-4) + 6(2-0) = 33$$

Sec 5.3

Cramer's rule, inverses, and volumes

Adjoint of A

Inverse from classical adjoints:

- **For any non-singular matrix A:**

$$A^{-1} = |A|^{-1} adj(A)$$

- **The classical adjoint (or adjugate) of A is the transpose of the matrix of cofactors:**

$$adj(A) = C^T$$

$$C = [c_{ij}], \qquad c_{ij} = (-1)^{(i+j)} |M_{ij}|$$

Example: 2 X 2 matrix

$$A^{-1} = |A|^{-1} adj(A),\ adj(A) = C^T,\ C = [c_{ij}],\ c_{ij} = (-1)^{(i+j)}|M_{ij}|$$

- **2x2:** $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$|A| = ad - bc$$

$$c_{11} = d, c_{12} = -c, c_{21} = -b, c_{22} = a \ \rightarrow C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

$$A^{-1} = |A|^{-1} C^T = \left(\frac{1}{ad - bc}\right) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example: 3 X 3 matrix

- **3x3:** $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \; A^{-1} = |A|^{-1} C^T$

$$\begin{aligned} |A| = \; & +a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ & -a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31} \end{aligned}$$

$$c_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, c_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{23} & a_{33} \end{vmatrix}, \cdots, c_{33} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

- **Nine 2x2 determinants! Not an efficient method.**
- **This is most useful for matrices with symbolic constants, rather than simply numbers.**

Cramer's Rule: Solve a linear system Ax = b

Given any non-singular square matrix A, we can directly write the solution for any right-hand side as:

$$A\vec{x} = \vec{b} \rightarrow \; x_i = \frac{|B_i|}{|A|}$$

- B_i **is the matrix obtained by replacing column** i **of** A **with** $\vec{b}$**:**

$$A = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{a}_i \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$
$$B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

Example: 2 X 2 matrix

$$A\vec{x} = \vec{b} \rightarrow \; x_i = \frac{|B_i|}{|A|}$$

- **2x2:** $\begin{matrix} ax & +by & = k \\ cx & +dy & = l \end{matrix} \rightarrow A\vec{x} = \vec{b}, A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \vec{b} = \begin{bmatrix} k \\ l \end{bmatrix}$

$$|A| = ad - bc$$

$$|B_1| = \begin{vmatrix} k & b \\ l & d \end{vmatrix} = kd - bl, \qquad |B_2| = \begin{vmatrix} a & k \\ c & l \end{vmatrix} = al - ck$$

$$x = x_1 = \frac{|B_1|}{|A|} = \frac{kd - bl}{ad - bc}, \qquad y = x_2 = \frac{|B_2|}{|A|} = \frac{al - ck}{ad - bc}$$

Example: 3 X 3 matrix

- **3x3:** $$A\vec{x} = \vec{b}, A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, x_i = \frac{|B_i|}{|A|}$$

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, |B_1| = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}, |B_2| = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, |B_3| = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

- **Four 3x3 determinants! Not an efficient method.**
- **Again, this is most useful for systems with symbolic constants, rather than simply numbers.**

Summary of Cramer's Rule

Given any non-singular square matrix A, we can directly write the solution for any right-hand side as:

$$A\vec{x} = \vec{b} \rightarrow \; x_i = \frac{|B_i|}{|A|}$$

- **For an $n \times n$ matrix, we need to compute $n + 1$ determinants, each with $n!$ terms of n elements:**

$$O(n^2 n!)$$

Applications: Area of Parallelogram

DEFINITION The *cross product* of $\boldsymbol{u} = (u_1, u_2, u_3)$ and $\boldsymbol{v} = (v_1, v_2, v_3)$ is a vector

$$\boldsymbol{u} \times \boldsymbol{v} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2v_3 - u_3v_2)\boldsymbol{i} + (u_3v_1 - u_1v_3)\boldsymbol{j} + (u_1v_2 - u_2v_1)\boldsymbol{k}. \tag{10}$$

This vector $\boldsymbol{u} \times \boldsymbol{v}$ is perpendicular to $\boldsymbol{u}$ and $\boldsymbol{v}$. The cross product $\boldsymbol{v} \times \boldsymbol{u}$ is $-(\boldsymbol{u} \times \boldsymbol{v})$.

$$\|\boldsymbol{u} \times \boldsymbol{v}\| = \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\,|\sin\theta| \qquad \text{and} \qquad |\boldsymbol{u} \cdot \boldsymbol{v}| = \|\boldsymbol{u}\|\,\|\boldsymbol{v}\|\,|\cos\theta|.$$

Note: Area of triangle = ½ ab sinθ. The length of the cross product is the area of the parallelogram which is twice of the area of the triangle.

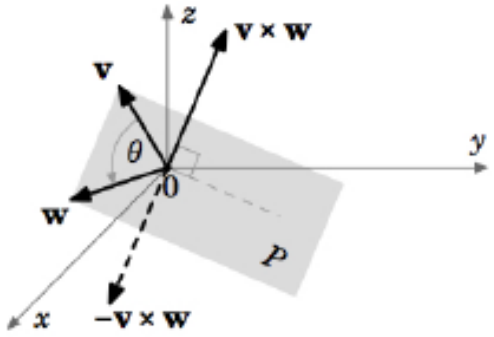


Figure 1.4.2 Direction of $\mathbf{v} \times \mathbf{w}$

We will now derive a formula for the magnitude of $\mathbf{v} \times \mathbf{w}$, for nonzero vectors $\mathbf{v}, \mathbf{w}$:

$$\begin{aligned}
\|\mathbf{v} \times \mathbf{w}\|^2 &= (v_2 w_3 - v_3 w_2)^2 + (v_3 w_1 - v_1 w_3)^2 + (v_1 w_2 - v_2 w_1)^2 \\
&= v_2^2 w_3^2 - 2 v_2 w_2 v_3 w_3 + v_3^2 w_2^2 + v_3^2 w_1^2 - 2 v_1 w_1 v_3 w_3 + v_1^2 w_3^2 + v_1^2 w_2^2 - 2 v_1 w_1 v_2 w_2 + v_2^2 w_1^2 \\
&= v_1^2 (w_2^2 + w_3^2) + v_2^2 (w_1^2 + w_3^2) + v_3^2 (w_1^2 + w_2^2) - 2(v_1 w_1 v_2 w_2 + v_1 w_1 v_3 w_3 + v_2 w_2 v_3 w_3)
\end{aligned}$$

and now adding and subtracting $v_1^2 w_1^2$, $v_2^2 w_2^2$, and $v_3^2 w_3^2$ on the right side gives

$$\begin{aligned}
&= v_1^2 (w_1^2 + w_2^2 + w_3^2) + v_2^2 (w_1^2 + w_2^2 + w_3^2) + v_3^2 (w_1^2 + w_2^2 + w_3^2) \\
&\quad - (v_1^2 w_1^2 + v_2^2 w_2^2 + v_3^2 w_3^2 + 2(v_1 w_1 v_2 w_2 + v_1 w_1 v_3 w_3 + v_2 w_2 v_3 w_3)) \\
&= (v_1^2 + v_2^2 + v_3^2)(w_1^2 + w_2^2 + w_3^2) \\
&\quad - ((v_1 w_1)^2 + (v_2 w_2)^2 + (v_3 w_3)^2 + 2(v_1 w_1)(v_2 w_2) + 2(v_1 w_1)(v_3 w_3) + 2(v_2 w_2)(v_3 w_3))
\end{aligned}$$

so using $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2ac + 2bc$ for the subtracted term gives

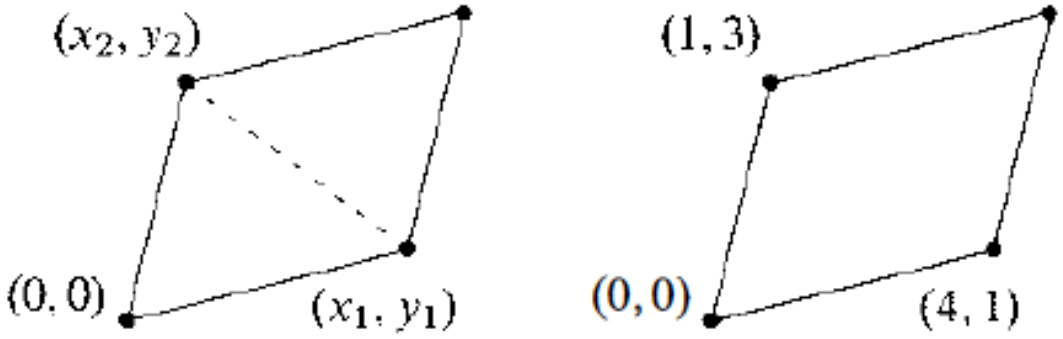
$$\begin{aligned}
&= (v_1^2 + v_2^2 + v_3^2)(w_1^2 + w_2^2 + w_3^2) - (v_1 w_1 + v_2 w_2 + v_3 w_3)^2 \\
&= \|\mathbf{v}\|^2 \, \|\mathbf{w}\|^2 - (\mathbf{v} \cdot \mathbf{w})^2 \\
&= \|\mathbf{v}\|^2 \, \|\mathbf{w}\|^2 \left(1 - \frac{(\mathbf{v} \cdot \mathbf{w})^2}{\|\mathbf{v}\|^2 \, \|\mathbf{w}\|^2}\right), \text{ since } \|\mathbf{v}\| > 0 \text{ and } \|\mathbf{w}\| > 0, \text{ so by Theorem 1.6} \\
&= \|\mathbf{v}\|^2 \, \|\mathbf{w}\|^2 (1 - \cos^2 \theta), \text{ where } \theta \text{ is the angle between } \mathbf{v} \text{ and } \mathbf{w}, \text{ so}
\end{aligned}$$

$$\|\mathbf{v} \times \mathbf{w}\|^2 = \|\mathbf{v}\|^2 \, \|\mathbf{w}\|^2 \, \sin^2 \theta, \text{ and since } 0^\circ \le \theta \le 180^\circ, \text{ then } \sin \theta \ge 0, \text{ so we have:}$$

Applications: Area of triangle

The triangle with corners (x_1, y_1) and (x_2, y_2) and (x_3, y_3) has **area** $= \dfrac{\textbf{determinant}}{\textbf{2}}$:

Area of triangle $\dfrac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$ $\quad$ Area $= \dfrac{1}{2}\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}$ when $(x_3, y_3) = (0, 0)$.



Parallelogram

Area $= \begin{vmatrix} 4 & 1 \\ 1 & 3 \end{vmatrix} = 11$

Triangle: Area $= \dfrac{\mathbf{11}}{\mathbf{2}}$

Figure 5.2: A triangle is half of a parallelogram. Area is half of a determinant.

Applications: Volume

Triple product $(\boldsymbol{u} \times \boldsymbol{v}) \cdot \boldsymbol{w} = \begin{vmatrix} w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$

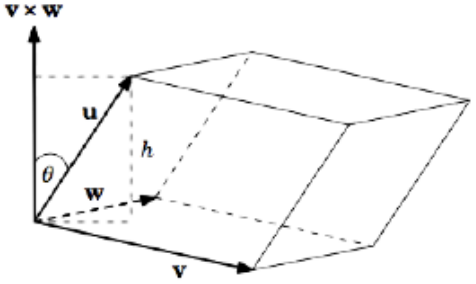


Figure 1.4.7 Parallelepiped P

Solution

Recall that the volume $\text{vol}(P)$ of a parallelepiped P is the area A of the base parallelogram times the height h. By Theorem 1.13 (b), the area A of the base parallelogram is $\|\mathbf{v} \times \mathbf{w}\|$. And we can see that since $\mathbf{v} \times \mathbf{w}$ is perpendicular to the base parallelogram determined by $\mathbf{v}$ and $\mathbf{w}$, then the height h is $\|\mathbf{u}\| \cos\theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v} \times \mathbf{w}$. By Theorem 1.6 we know that

$$\cos\theta = \frac{\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})}{\|\mathbf{u}\| \, \|\mathbf{v} \times \mathbf{w}\|}. \text{ Hence,}$$

$$\begin{aligned} \text{vol}(P) &= A\,h \\ &= \|\mathbf{v} \times \mathbf{w}\| \, \frac{\|\mathbf{u}\| \, \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})}{\|\mathbf{u}\| \, \|\mathbf{v} \times \mathbf{w}\|} \\ &= \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) \end{aligned}$$

Summary:

- **The determinant provides a tools that allows us to write analytical expressions for:**
  - **The singularity of a matrix.**
  - **The inverse of nonsingular matrix.**
  - **The solution to a system with a nonsingular coefficient matrix.**
- **This method is not well-suited to solve general systems:**
  - **Elimination is much faster for all but the smallest systems.**

Examples

Examples using determinant.

Ex 1: Determine the conditions on singularity.

- **For what values of a, b, c and d is the following matrix singular?**

$$A = \begin{bmatrix} 1 & a & 1 \\ b & 1 & c \\ 1 & d & 1 \end{bmatrix}$$

Conditions on singularity.

$$\begin{vmatrix} 1 & a & 1 \\ b & 1 & c \\ 1 & d & 1 \end{vmatrix} = 1\begin{vmatrix} 1 & c \\ d & 1 \end{vmatrix} - a\begin{vmatrix} b & c \\ 1 & 1 \end{vmatrix} + 1\begin{vmatrix} b & 1 \\ 1 & d \end{vmatrix}$$

$$= (1 - cd) - a(b - c) + (bd - 1)$$

$$= 1 - cd - ab + ac + bd - 1$$

$$= a(c - b) + d(b - c) = (a - d)(c - b)$$

$$|A| = 0 \rightarrow (a - d)(c - d) = 0 \rightarrow a = d \text{ or } b = c$$

- A **will be singular if either** $a = d$ **or** $b = c$.

Ex 2: Cramer's Rule

- **Find a general form for the solution of the following system of equations.**

$$\begin{aligned} x_1 &= & ax_2 & +ax_3 & +b \\ x_2 &= ax_1 & & +ax_3 & +b \\ x_3 &= ax_1 & +ax_2 & & +b \end{aligned}$$

- **Recall:** $x = \frac{|B|}{|A|}, B = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_{i-1} & \vec{b} & \vec{a}_{i+1} & \cdots & \vec{a}_n \end{bmatrix}$

Cramer's Rule

- **Recast in matrix form:**

$$\begin{aligned} x_1 &= x_2 + ax_3 + b \\ x_2 &= ax_1 + ax_3 + b \\ x_3 &= ax_1 + ax_2 + b \end{aligned} \rightarrow \boxed{\vec{x} = A\vec{x} + \vec{b} \rightarrow (I - A)\vec{x} = \vec{b}}$$

$$A = \begin{bmatrix} 0 & a & a \\ a & 0 & a \\ a & a & 0 \end{bmatrix}, M = (I - A) = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, \quad x_i = \frac{|B_i|}{|M|}, B_i = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_{i-1} & \vec{b} & \vec{a}_{i+1} & \cdots & \vec{a}_n \end{bmatrix}$$

$$|M| = 1\begin{vmatrix} 1 & -a \\ -a & 1 \end{vmatrix} + a\begin{vmatrix} -a & -a \\ -a & 1 \end{vmatrix} - a\begin{vmatrix} -a & 1 \\ -a & -a \end{vmatrix}$$

$$= \left(1 - a^2\right) + a\left(-a - a^2\right) - a(a^2 + a)$$

$$= 1 - a^2 - a^2 - a^3 - a^3 - a^2$$

$$= 1 - 3a^2 - 2a^3 = (1 + a)\left(1 - a - 2a^2\right)$$

$$= (1 + a)(1 + a)(1 - 2a) = (1 + a)^2(1 - 2a)$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix},$$

$$x_i = \frac{|B_i|}{|M|}, B_i = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_{i-1} & \vec{b} & \vec{a}_{i+1} & \cdots & \vec{a}_n \end{bmatrix}$$

$$|B_1| = \begin{vmatrix} b & -a & -a \\ b & 1 & -a \\ b & -a & 1 \end{vmatrix} = b \begin{vmatrix} 1 & -a \\ -a & 1 \end{vmatrix} + a \begin{vmatrix} b & -a \\ b & 1 \end{vmatrix} - a \begin{vmatrix} b & 1 \\ b & -a \end{vmatrix}$$

$$= b\left(1 - a^2\right) + a(b + ab) - a(-ab - b)$$

$$= b - ba^2 + ab + a^2b + a^2b + ab$$

$$= b + 2ab + a^2b = b\left(1 + 2a + a^2\right) = b(1 + a)^2$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, \quad x_i = \frac{|B_i|}{|M|}, B_i = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_{i-1} & \vec{b} & \vec{a}_{i+1} & \cdots & \vec{a}_n \end{bmatrix}$$

$$|B_2| = \begin{vmatrix} 1 & b & -a \\ -a & b & -a \\ -a & b & 1 \end{vmatrix} = 1 \begin{vmatrix} b & -a \\ b & 1 \end{vmatrix} - b \begin{vmatrix} -a & -a \\ -a & 1 \end{vmatrix} - a \begin{vmatrix} -a & b \\ -a & b \end{vmatrix}$$

$$= (b + ab) - b\left(-a - a^2\right) - a(-ab + ab)$$

$$= b + ab + ab + a^2 b$$

$$= b + 2ab + a^2 b = b(1 + a)^2$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix},$$

$$x_i = \frac{|B_i|}{|M|}, B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

$$|B_3| = \begin{vmatrix} 1 & -a & b \\ -a & 1 & b \\ -a & -a & b \end{vmatrix} = 1\begin{vmatrix} 1 & b \\ -a & b \end{vmatrix} + a\begin{vmatrix} -a & b \\ -a & b \end{vmatrix} + b\begin{vmatrix} -a & 1 \\ -a & -a \end{vmatrix}$$

$$= (b + ab) + a(-ab + ab) + b(a^2 + a)$$

$$= b + ab + a^2b + ab$$

$$= b + 2ab + a^2b = b(1 + a)^2$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, \quad x_i = \frac{|B_i|}{|M|}, B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

$$|M| = (1+a)^2\,(1-2a)$$

$$|B_1| = |B_2| = |B_3| = b(1+a)^2$$

$$x_1 = x_2 = x_3 = \frac{b(1+a)^2}{(1+a)^2(1-2a)} = \left(\frac{b}{1-2a}\right), a \neq -1, \frac{1}{2}$$

Ex 4: Inverse by classical adjoint.

- **Find a general form for the inverse of the following matrix:**

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}$$

- **Recall:** $A^{-1} = |A|^{-1} C^T, C = [c_i\ ],\ c_{ij} = (-1)^{(i+j)} |M_{ij}|$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}, A^{-1} = |A|^{-1}C^T, C = [c_{i\,j}], \qquad c_{ij} = (-1)^{(i+j)}\,|M_{ij}|$$

$$|A| = a\begin{vmatrix} 0 & b \\ b & b \end{vmatrix} - a\begin{vmatrix} a & b \\ 0 & b \end{vmatrix} + 0\begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix}$$

$$= a\left(0 - b^2\right) - a(ab - 0) + 0$$

$$= 0 - ab^2 - a^2b + 0$$

$$= -ab(a + b)$$

$$|A| \neq 0 \rightarrow a, b \neq 0, a \neq -b$$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}, A^{-1} = |A|^{-1} C^T, C = [c_{ij}], \qquad c_{ij} = (-1)^{(i+j)} \, |M_{ij}|$$

$$c_{11} = + \begin{vmatrix} 0 & b \\ b & b \end{vmatrix} = -b^2, c_{12} = - \begin{vmatrix} a & b \\ 0 & b \end{vmatrix} = -ab, c_{13} = + \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab$$

$$c_{21} = - \begin{vmatrix} a & 0 \\ b & b \end{vmatrix} = -ab, c_{22} = + \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab, c_{23} = - \begin{vmatrix} a & a \\ 0 & b \end{vmatrix} = -ab$$

$$c_{31} = + \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab, c_{32} = - \begin{vmatrix} a & 0 \\ a & b \end{vmatrix} = -ab, c_{33} = + \begin{vmatrix} a & a \\ a & 0 \end{vmatrix} = -a^2$$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}, A^{-1} = |A|^{-1}C^T, C = [c_{i\,j}], \qquad c_{ij} = (-1)^{(i+j)}\,|M_{ij}|$$

$$|A| = -ab(a+b)$$

$$\begin{matrix} c_{11} = -b^2 & c_{12} = -ab & c_{13} = ab \\ c_{21} = -ab & c_{22} = ab & c_{23} = -ab \\ c_{31} = ab & c_{32} = -ab & c_{33} = -a^2 \end{matrix} \rightarrow C = \begin{bmatrix} -b^2 & -ab & ab \\ -ab & ab & -ab \\ ab & -ab & -a^2 \end{bmatrix}$$

$$A^{-1} = |A|^{-1}C^T = \left(\frac{1}{-ab(a+b)}\right)\begin{bmatrix} -b^2 & -ab & ab \\ -ab & ab & -ab \\ ab & -ab & -a^2 \end{bmatrix}$$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}$$

$$A^{-1} = \left(\frac{1}{-ab(a+b)}\right)\begin{bmatrix} -b^2 & -ab & ab \\ -ab & ab & -ab \\ ab & -ab & -a^2 \end{bmatrix}$$

$$A^{-1} = \left(\frac{1}{a+b}\right)\begin{bmatrix} b/a & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & a/b \end{bmatrix}, a, b \neq 0, a \neq -b$$